

Optimal All-to-All Broadcast in WDM Optical Networks with Breakdown or Power-Off Transceivers *

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Abstract

Packet scheduling provides a way to reduce routing control and guarantee overall throughput. An interesting special case of the packet scheduling is all-to-all broadcast in which every transmitter/receiver pair has exactly one packet to be transferred. Several optimal scheduling algorithms have been proposed for the all-to-all broadcast problem in Wavelength Division Multiplexed (WDM) optical star networks. However, these algorithms do not fit anymore when failures occur in some transceivers. In this paper, we consider the all-to-all broadcast problem in WDM optical star network with some breakdown or power-off transceivers. We focus on the all-optical scheduling problem where the traffic reaches its destination in single-hop without being converted to electronic form for reaching high data transmission rates. The transmitters are tunable with an associated tuning delay, and the receivers are fixed-tuned to one of the available wavelengths. In this model, we first identify the lower bounds of the scheduling length for the problem. Then, we propose single-hop scheduling algorithms to find the optimal solution in both terms of arbitrary number of wavelengths and value of tuning latency.

Wavelength division multiplexing (WDM) is one of the techniques widely used to exploit the huge optical bandwidth of the fiber. The WDM technique provides concurrent transmissions on an optical fiber through different channels of wavelengths. The broadcast-star topology is one of the most popular packet switched architecture in WDM networks because of its simplicity [1].

In general, network transceivers can be classified into four categories depending on the tunability of transceivers for each node: fixed transmitter(s) and fixed receiver(s) (FT-FR); fixed transmitter(s) and tunable receiver(s) (FT-TR); tunable transmitter(s) and fixed receiver(s) (TT-FR); and tunable transmitter(s) and tunable receiver(s) (TT-TR). However, the tunability is an expensive alternative, hence each node is usually equipped with one tunable transmitter and one fixed receiver (TT/FR) [4]. Each tunable transmitter has tuning latency δ to tune from one wavelength to another that will affect the performance of scheduling algorithms because the tuning latency is greater than the packet transmission time. Tuning latency limits the network throughput as nodes spend time in tuning which could otherwise be used in useful information transfer. Therefore, how to minimize the effect of tuning latency on the transmission scheduling for a given traffic matrix is a critical issue.

The transmission scheduling methods are generally classified into two categories: single-hop and multi-hop [5, 6]. The terms single-hop and multi-hop indicate whether user data only traverses optical switching components on the end-to-end path (single-hop) or whether it traverses a combination of optical and electronic switching components (multi-hop). Obviously, maintaining the

1 Introduction

Optical networks provide higher transmission quality and larger bandwidth for high-speed communication than traditional electronic networks.

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data signal in single-hop will allow the all-optical transmission and eliminate the electronic bottleneck of electronic switching components. We are concerned with the single-hop scheduling where the traffic reaches its destination without being converted to electronic form for reaching high data transmission rate. In an N -node WDM broadcast-star network, the single-hop packet transmission scheduling problem can be reduced to the open shop scheduling problem when tuning latency $\delta = 0$ [7]. Packet transmission is called non-preemptive if the transmitter is not involved in any other transmission until it completes its current transmission and preemptive otherwise. The non-preemptive open shop scheduling problem is NP-Complete when $N \geq 3$ and can be solved in polynomial time if $N = 2$ [8]. However, the single-hop packet transmission scheduling problem has been proven as NP-Complete even when $N = 2$ [7].

In this paper, we consider only on an interesting special case of the single-hop scheduling which is called all-to-all broadcast [9, 10, 13]. Every node pair has exactly one packet to be transferred for the all-to-all broadcast. If each node transmits packets to all nodes including itself, N^2 packets should be transmitted for an N -node WDM broadcast-star network. In [9], Pieris and Sasaki proposed several upper and lower bounds of minimum scheduling length for the all-to-all broadcast problem. They assumed that the transmitters are tunable to any one of wavelengths with tuning latency δ , and the the number of available wavelengths is large enough. If packet transmissions are allowed while a transmitter or receiver is tuning from one wavelength to another, they found that the upper and lower bounds for the minimum scheduling length of all-to-all broadcast are $(N + o(N)) * (\sqrt{\delta} + 1)$ and $(N + o(N)) * \sqrt{\delta}$, respectively. However, there is a tune-transmit separability constraint that packets can not be transmitted while a transmitter or receiver is tuning. For satisfying the tune-transmit separability constraint, they found that the upper and lower bounds for the minimum scheduling length of all-to-all broadcast are $(N + o(N)) * (\lceil \sqrt{\delta} \rceil + \sqrt{\delta})$ and $(2N + o(N)) * \sqrt{\delta}$, respectively.

In [10], Choi, Choi and Azizoglu considered an important limitation when the number of wavelengths w is significantly less than the number of transceivers N , and assumed that each wavelength is exactly shared by N/w receivers for a uniform loading of wavelengths. They established Lower bounds on the minimum scheduling length to ex-

change packets between every node pair in a network with tunable transmitters and fixed-tuned receivers. They showed that the lower bound for the minimum scheduling length of all-to-all broadcast is at least $\max\{\delta + \frac{N^2}{w}, w\delta + \frac{N^2}{w^2} + N - \frac{N}{w}\}$, and the upper bound obtained in [9] is in fact a lower bound to the optimum schedule length, i.e., the all-to-all transmission schedule presented in [9] is an optimal schedule for the case of uniform loading of wavelengths. In [11], they also consider the case when N is not divisible by w . The wavelengths are loaded with fixed-tuned receivers as uniformly as possible. They proposed optimal scheduling algorithms for three different cases depending on the range of tuning latency δ . Excepting for a very narrow range, their algorithms perform within a factor $13/12$ of the lower bound.

If each node transmits packets to all nodes excepting itself, the number of packets to be sent for the all-to-all broadcast is $N(N - 1)$. In [12], Lee, Oh and Choi showed that $\max\{\frac{N(N-1)}{k}, w\delta + N - 1\}$ is a lower bound for the cycle lengths of schedules and proposed an algorithm for obtaining a transmission schedule that meets this lower bound under the assumption that N is divisible by w . In [13], Yeh, Wee and Hong studied both kinds of all-to-all connection problems depending on whether each node transmits packets to all nodes including or excepting itself and proposed two lower bounds for these two kinds of problems, $\max\{\lceil \frac{N}{w} \rceil N, w\delta + N\}$ and $\max\{\lceil \frac{N}{w} \rceil (N - 1), w\delta + N - 1\}$, respectively. They also proposed scheduling algorithms to find the optimal scheduling for arbitrary values of N , w , and δ .

Choi, Choi and Ni [14] studied a case when transmitters may access to only a limited number of channels, i.e., the number of channels to which a laser can tune may be much smaller than the number of channels. They developed an algorithm that provides an optimal for all-to-all broadcast with limited tuning ranges. In [15], Lee, Oh and Choi considered the problem as a virtual topology embedding problem. If each node is equipped with α tunable transmitters and β fixed-tuned receivers, they developed an optimal transmission schedule with schedule length $\max\{\alpha\beta N(N-1)/w, \beta(N-1) + w\delta\}$ such that each transmitter of each node transmits once to each receiver of its neighboring nodes within this time period. In [16], Li conducted probabilistic analysis of the average-case performance of a packet transmission scheduling algorithm for the cyclic all-to-all broadcast. Their result implies that by using currently available tunable optical transceivers, it

is possible to build single-hop WDM networks that efficiently utilize all the wavelengths.

In this paper, we consider the all-to-all broadcast problem in WDM optical star network with some breakdown or power-off transceivers. Each node in the network is assigned a tunable transmitter and a fixed-tuned receiver. We identify the lower bounds of the minimum scheduling length for the problem and propose single-hop scheduling algorithms to find the optimal solution in both terms of arbitrary number of wavelengths and value of tuning latency.

2 Problem Definition

This paper considers all-to-all connection problem in WDM optical passive star networks with N nodes and w wavelengths. We assume that each node is equipped with exactly one tunable transmitter and one fixed receiver (TT-FR), and each transmitter can tune to all available wavelengths with the tuning latency δ which is expressed in units of packet durations. A single-hop scheduling protocol is mandatory so that each packet can directly be transmitted to its destination node in a unit time slot. Our objective is to schedule the packet transmissions for the all-to-all broadcast in order to minimize the scheduling length.

Since the number of wavelengths is significantly less than the number of nodes, whether N is divisible by w or not, we focus on the case when some nodes crash and the received wavelengths are not reassigned to all nodes in that they are fixed. For example, a WDM network consisting of 12 nodes is considered in Figure 1(a). When 4 nodes fail, the remaining 8 nodes is indicated in Figure 1(b) in which each group consists of different number of nodes. The previous optimal scheduling does not fit in with the remaining networks anymore and we need to redesign the scheduling algorithms to minimize the scheduling length.

We assume that there are B damaged nodes occurring in the network in which the remaining $N - B$ nodes are represented by $n_0, n_1, \dots, n_{N-B-1}$ with the same w available wavelengths on each link represented by $\lambda_0, \lambda_1, \dots, \lambda_{w-1}$. Let FR-group denote a group of the nodes using the same received wavelength for transmissions, G_i denote an FR-group, δ denote the non-negligible tuning time of each tuning step, and λ_i denote the received wavelength used by each node in G_i . Therefore, the remaining $N - B$ nodes are partitioned into w FR-groups represented by $G_0, G_1, \dots$, and G_{w-1} .

3 Lower Bounds of the Scheduling Length

In this section, we identify some lower bounds for both kinds of all-to-all connection scheduling problems and show that the scheduling length found by the algorithm we propose meet these lower bounds. In all-to-all connection scheduling, the packet transmissions is scheduled for each nodes-pair so that all nodes transmit only once to each node within a repeating cycle time as less as possible.

There are two different kinds of all-to-all connection problems depending on that the nodes send packets to all nodes including or except itself.

In the case when each node sends packets to all nodes including itself, there are $(N - B)^2$ packet transmissions in the scheduling. A theorem follows:

Theorem 3.1. *The lower bound of the scheduling length of any all-to-all connection scheduling is $\max\{(\max |G_i|)(N - B), w\delta + N - B\}$, where the nodes send the packets to all nodes including themselves.*

Proof. First, each node transmits $N - B$ packets by w available wavelengths in the scheduling. If we take into consideration that all-to-all connection are repeated periodically, each node has to tune its transmitter w times. For considering only one node, the scheduling length is at least the sum of total tuning latency and transmission time for $N - B$ packets, i.e., $w\delta + N - B$. Second, $N - B$ packets must be received by only one wavelength fixed to its receiver of each node in each FR-group. These packets are sent to all nodes including itself. Since no more than two packets can be sent using the same wavelength at the same time, the scheduling length is at least the number of packets, $|G_i|(N - B)$ for $0 \leq i \leq w$, and the packet is transmitted using a wavelength λ_i . Since the size of each group is different, the scheduling length is at least $(\max |G_i|)(N - B)$. $\square$

In the case when each node sends packets to all nodes excepting itself, each of $(N - B)(N - B - 1)$ node-pairs has a single packet to send in the scheduling. The number of packets sent to a group of nodes, G_i , using the wavelength λ_i can be different if the transmitting node is included in G_i where $0 \leq i \leq w$. Hence, the lower bound of optimal scheduling length is slightly changed. A theorem follows:

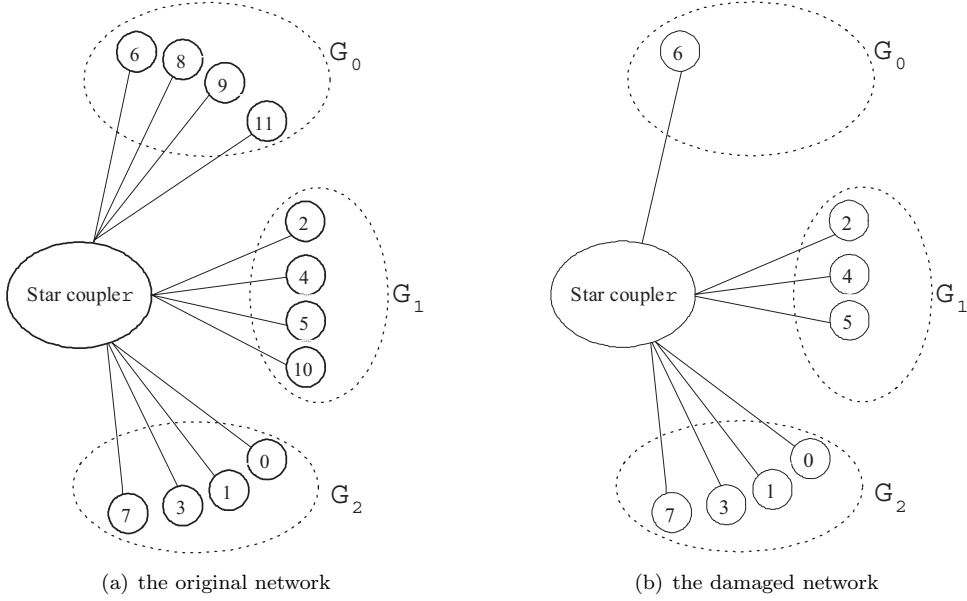


Figure 1: WDM optical passive star networks with 12 nodes and 8 nodes respectively

Theorem 3.2. *The lower bound of the scheduling length of any all-to-all connection scheduling is $\max\{(\max |G_i|)(N - B - 1), w\delta + N - B - 1\}$, where the nodes transmit the packets to all nodes except themselves.*

Proof. The proof is the same manner as in Theorem 3.1. $\square$

4 Optimal Single-hop Scheduling Algorithms

A schedule of transmissions is like a timetable that arranges each packet being sent at which time by which wavelength. In order to send a packet, the transmitter must tune to the received wavelength and it cannot transmit packets while it is tuning to another wavelength. Hence, it transmits all packets that use the same wavelength without interruption if possible and then tunes to another wavelength for another transmission. Also a wavelength can be used many times for packet transmissions because the number of wavelengths is much less than that of nodes in WDM networks. Thus, an optimal scheduling can be found based on pipelining the packets transmissions. All the scheduling algorithms proposed in the previous research works are to tune the available wavelengths to the received nodes. In our algorithms, the received wavelengths is fixed and we only consider

the scheduling steps for packet transmissions. In what follows, we present two optimal scheduling algorithms for two kinds of the all-to-all broadcast and two optimal scheduling algorithms for a special case.

4.1 Optimal Scheduling Algorithm for Sending Packets to Themselves

We consider the all-to-all connection problem in which each node sends the packet to itself only once. Since the size of FR-group is different from that of others, we first sort the size in descending order as: $\psi_0, \psi_1, \dots$, and ψ_{w-1} , where $|\psi_i| \geq |\psi_j|$ if $j \geq i$, $0 \leq i, j \leq w-1$. For example, as indicated in Figure 1(b), $\psi_0 = \{0, 1, 3, 7\}$, $\psi_1 = \{2, 4, 5\}$, and $\psi_2 = \{6\}$. We aim at that each node in the largest FR-group sends the packets to other nodes in the smallest FR-group. Let n_j^i be the j -th node of ψ_i , where $0 \leq j \leq |\psi_i| - 1$. Each node n_p starts transmitting packets to the nodes in group ψ_i right after finishing transmitting packets to the nodes in group ψ_{i-1} only with the tuning delay δ . The time s_p when node n_p transmits the every first packets is represented as follows: $s_p = |\psi_0|p + 1, 0 \leq p \leq N - B - 1$.

In what follows, we introduce *Algorithm1* for the all-to-all connection problems. *Algorithm1* shows the transmission scheduling settling tuning delay and is lack of the wavelengths. In *Algorithm1*, $T(n_p, t) = n_j^i$ is that the node n_p

transmits a packet to the j -th node of ψ_i at time t . And the scheduling length of optimal scheduling L_{opt} is defined by $\max\{|\psi_0|(N-B), w\delta + N-B\}$.

Algorithm1

- 1: Sort FR-groups in the descending order of group sizes and the sorted order is: $\psi_0, \psi_1, \dots$, and ψ_{w-1} , where $|\psi_i| \geq |\psi_j|$ if $j \geq i$, $0 \leq i, j \leq w-1$. Let n_j^i be the j -th node of ψ_i , where $0 \leq j \leq |\psi_i| - 1$.
- 2: For $p = 0$ to $N - B - 1$ do
- 3: $t \leftarrow s_p$
- 4: For $i = 0$ to $w - 1$ do
- 5: For $j = 0$ to $|\psi_i| - 1$ do
- 6: $T(n_p, t) \leftarrow n_j^i$
- 7: $t \leftarrow t + 1$
- 8: $t \leftarrow t + \delta$

Figure 2 shows a timetable of transmission scheduling constructed by *Algorithm1* if $N - B = 8$, $w = 3$, and $\delta = 5$. The numbers on the leftmost column represent the nodes and time goes on from left to right. In what follows, we will prove that *Algorithm1* can find a correct scheduling length for all-to-all connection problems.

Lemma 4.1. *No two nodes apply the same wavelength for transmitting at the same time in the scheduling generated by Algorithm1.*

Proof. We only have to verify that the two nodes n_p and n_{p+1} do not use the same transmitting wavelength at the same time. Let α be the time when the node n_p finishes transmitting packets for the group ψ_i and β be the time when node n_{p+1} finishes tuning its transmitter to start the packet transmitting using the received wavelength of group ψ_i , α and β can be represented as follows: $\alpha = s_p + \text{the number of packets transmitted} + \text{the total tuning delay} = (|\psi_0|p + 1) + (\sum_{k=0}^i |\psi_k|) + i\delta$.

Similarly, $\beta = (|\psi_0|(p+1) + 1) + (\sum_{k=0}^{i-1} |\psi_k|) + i\delta$. We obtain that $\beta - \alpha = |\psi_0| - |\psi_i|$. Hence, $|\psi_0| \geq |\psi_i|$, $0 \leq i \leq w-1$, $\beta \geq \alpha$ always holds for any value p . $\square$

Lemma 4.2. *In the scheduling constructed by Algorithm1, the time L_p for each node n_p to transmit all packets for the all-to-all connection is equal to $(w-1)\delta + N - B$ for any p in $0 \leq p \leq N - B - 1$.*

Proof. From the above lemma, we assure that each node does not have to wait for using a wavelength which is used by another node. In other words, each node can start transmitting packets as soon as tuning to the wavelength is completed. No doubt, every node n_p transmits $N - B - 1$ packets in the scheduling generated by *Algorithm1*. Clearly, each node n_p will tune $w - 1$ times, so $L_p = (w - 1)\delta + N - B$ for any p . $\square$

The scheduling generated by *Algorithm1* have the characteristic that all L_p are identical for $0 \leq p \leq N - B - 1$, so we denote L_p simply as L . Now, we repeat the transmission scheduling constructed by *Algorithm1* until we obtain that any time span of L_{opt} starting from any position of the repeated scheduling is an optimal scheduling. *Algorithm2* will find an optimal scheduling by repeating *Algorithm1*. *Algorithm2* differs from *Algorithm1* in that the time t is repeated by $t - L_{opt}$ if t is greater than L_{opt} .

Algorithm2

- 1: Sort FR-groups in the descending order of group sizes and the sorted order is: $\psi_0, \psi_1, \dots$, and ψ_{w-1} , where $|\psi_i| \geq |\psi_j|$ if $j \geq i$, $0 \leq i, j \leq w-1$. Let n_j^i be the j -th node of ψ_i , where $0 \leq j \leq |\psi_i| - 1$.
- 2: For $p = 0$ to $N - B - 1$ do
- 3: $t \leftarrow s_p$
- 4: For $i = 0$ to $w - 1$ do
- 5: For $j = 0$ to $|\psi_i| - 1$ do
- 6: if $(t > L_{opt})$ then $T(n_p, t - L_{opt}) \leftarrow n_j^i$
- 7: else $T(n_p, t) \leftarrow n_j^i$
- 9: $t \leftarrow t + 1$
- 9: $t \leftarrow t + \delta$

Figure 3 shows the repetitions of transmission scheduling of Figure 2. Note that in Figure 3, $L_{opt} = 32$ is calculated from N , B , w , and δ . If we take out any portion of Figure 3 and span 32 time units, an optimal scheduling is found as indicated in Figure 4.

Theorem 4.3. *For any values w and δ , the scheduling constructed by Algorithm2 is an optimal scheduling.*

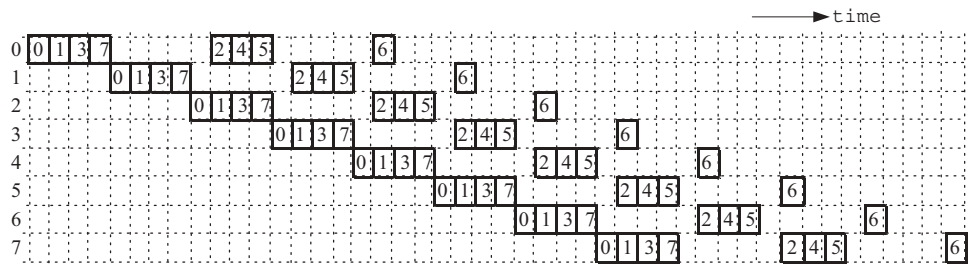


Figure 2: An Example of *Algorithm1*

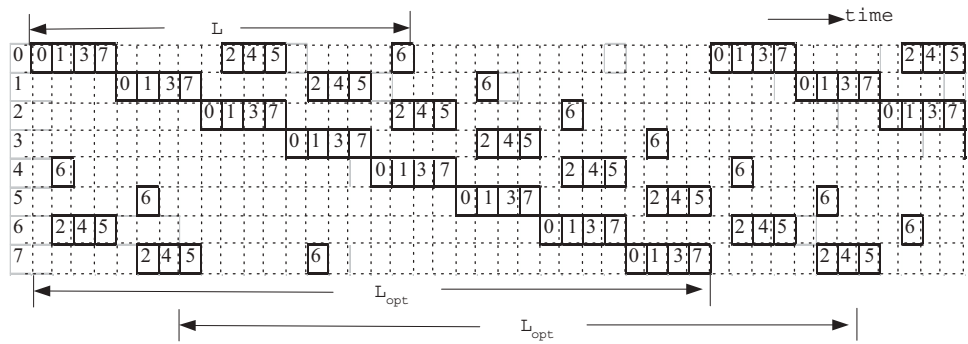


Figure 3: An Example of Repeating *Algorithm1*.

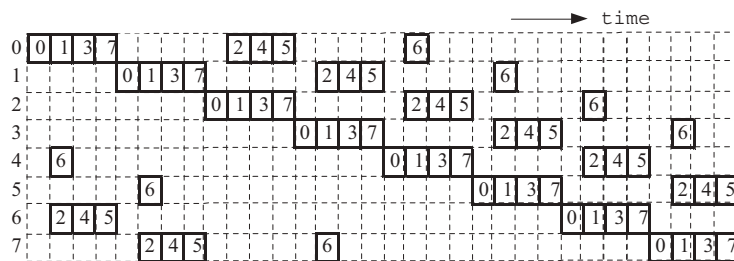


Figure 4: An Example of Optimal scheuling with $(N - B)^2 = 64$ Pakets.

Proof. Now, we must prove that $L_{opt} - L$ is no less than tuning latency δ in *Algorithm2*. From Lemma 4.2, $L = (w-1)\delta + N - B$. If L_{opt} is equal to $|\psi_0|(N-B)$, then $|\psi_0|(N-B) \geq w\delta + N - B \Rightarrow |\psi_0|(N-B) - [(w-1)\delta + N - B] \geq \delta \Rightarrow L_{opt} - L \geq \delta$.

Otherwise, we choose that L_{opt} is equal to $w\delta + N - B$, so $L_{opt} - L = w\delta + N - B - [(w-1)\delta + N - B] = \delta$. Thus, no matter L_{opt} is equal to either $|\psi_0|(N-B)$ or $w\delta + N - B$, we prove $L_{opt} - L \geq \delta$. $\square$

Figure 4 shows an optimal scheduling which is $|\psi_0|(N-B) = 4 \times 8$. Clearly, the scheduling of Figure 4 is not affected by the tuning latency $\delta = 5$. Now, we give an example $N - B = 8$, $w = 3$, and $\delta = 9$, and the tuning latency will determine the scheduling length by *Algorithm2*. Figure 5 shows the optimal scheduling in which the scheduling length is $w\delta + N - B = 3 \times 9 + 8 = 35$.

4.2 Optimal Scheduling Algorithm for Not Sending Packets to Themselves

The other scheduling problem for scheduling all-to-all broadcast is that every node can not send any packet to itself. This case is more complicated than the above case. Some notations is similar to the above definitions. The time s'_p when node n_p transmits the every first packet is set as follows: $s'_p = |\psi_0|p - |\psi_0 \cap \{0, 1, 2, \dots, p-1\}| + 1$

In the following algorithms, let $T(n_p, t) = n_j^i$ denote that node n_p transmits a packet to the j -th node of ψ_i at time t . And the scheduling length of optimal scheduling L_{opt} is changed to $\max\{|\psi_0|(N-B-1), w\delta + N - B - 1\}$.

The case is more difficult than the case which nodes can send to themselves. Therefore, we can adopt *Algorithm2* for this case. The scheduling length created by *Algorithm2* is at most $\max\{|\psi_0|(N-B), w\delta + N - B\}$, which these constructed scheduling are not optimal. In what follows, we will discuss a special case which satisfies that $|\psi_0| > |\psi_i|$, where $1 \leq i \leq w-1$. *Algorithm3* finds an optimal scheduling for this special case and is described as follows:

Algorithm3

- 1: Sort FR-groups in the descending order of group sizes and the sorted order is: $\psi_0, \psi_1, \dots$, and ψ_{w-1} , where $|\psi_0| > |\psi_i|$, and $|\psi_i| \geq |\psi_j|$ if $j \geq i$, $1 \leq i, j \leq w-1$. And Let n_j^i denote the j -th node of ψ_i , where $0 \leq j \leq |\psi_i| - 1$.

- 2: For $p = 0$ to $N - B - 1$ do
- 3: $t \leftarrow s'_p$
- 4: For $i = 0$ to $w - 1$ do
- 5: For $j = 0$ to $|\psi_i| - 1$ do
- 6: if $(n_p \neq n_j^i)$
- 7: $T(n_p, t) \leftarrow n_j^i$
- 8: $t \leftarrow t + 1$
- 9: $t \leftarrow t + \delta$

Algorithm3 is similar to *Algorithm1*. The difference of these two algorithms is in the line 6 of *Algorithm3* which renders that each node can not transmit packets to itself. Figure 6 shows a timetable of transmission scheduling constructed by *Algorithm3* if $N - B = 8$, $w = 3$, and $\delta = 5$. In what follows, We will prove that *Algorithm3* can find a suitable scheduling for this special case.

Lemma 4.4. *No two nodes using the same wavelength for transmitting at the same time in the scheduling generated by Algorithm3.*

Proof. We also verify that the two nodes n_p and n_{p+1} do not use the same transmitting wavelength at the same time. Let α denote the time when the node n_p finishes transmitting packets to the j -th node of group ψ_i , where $0 \leq j \leq |\psi_i| - 1$ and β be the time when node n_{p+1} finishes tuning its transmitter to star packet transmissions using the received wavelength of group ψ_i , α and β can be represented as follows: $\alpha = s_p +$ the number of packets transmitted + the total tuning delay = $(|\psi_0|p - |\psi_0 \cap \{0, 1, 2, \dots, p-1\}| + 1) + (\sum_{k=0}^i |\psi_k| - |\{p\} \cap \bigcup_{k=0}^i \psi_k|) + i\delta$.

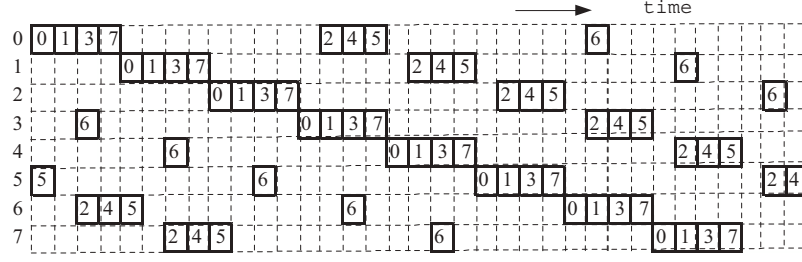
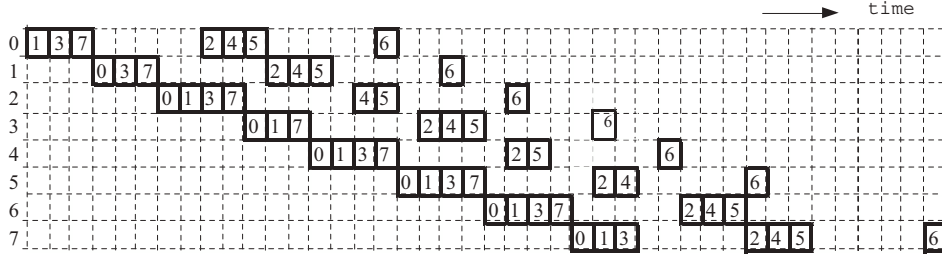
Similarly, $\beta = (|\psi_0|(p+1) - |\psi_0 \cap \{0, 1, 2, \dots, p\}| + 1) + (\sum_{k=0}^{i-1} |\psi_k| - |\{p+1\} \cap \bigcup_{k=0}^{i-1} \psi_k|) + i\delta$.

We obtain that $\beta - \alpha = |\psi_0| - (|\psi_0 \cap \{0, 1, 2, \dots, p\}| - |\psi_0 \cap \{0, 1, 2, \dots, p-1\}|) - |\psi_i| + (|\{p\} \cap \bigcup_{k=0}^i \psi_k| - |\{p+1\} \cap \bigcup_{k=0}^{i-1} \psi_k|)$.

If ψ_0 contains both n_p and n_{p+1} , then $(|\psi_0 \cap \{0, 1, 2, \dots, p\}| - |\psi_0 \cap \{0, 1, 2, \dots, p-1\}|) = 0$ and $(|\{p\} \cap \bigcup_{k=0}^i \psi_k| - |\{p+1\} \cap \bigcup_{k=0}^{i-1} \psi_k|) = 0$. Since $|\psi_0| > |\psi_i|$, $\beta - \alpha = |\psi_0| - |\psi_i| > 0$.

When ψ_0 includes n_{p+1} but not n_p , $(|\psi_0 \cap \{0, 1, 2, \dots, p\}| - |\psi_0 \cap \{0, 1, 2, \dots, p-1\}|) = -1$, and $(|\{p\} \cap \bigcup_{k=0}^i \psi_k| - |\{p+1\} \cap \bigcup_{k=0}^{i-1} \psi_k|) \geq -1$. We obtain $\beta - \alpha \geq |\psi_0| - |\psi_i| > 0$.

When ψ_0 includes n_p but not n_{p+1} , $(|\psi_0 \cap \{0, 1, 2, \dots, p\}| - |\psi_0 \cap \{0, 1, 2, \dots, p-1\}|) = 1$,


 Figure 5: An Example of Optimal Scheuling for $N - B = 8$, $w = 3$, and $\delta = 9$.

 Figure 6: An Example of *Algorithm3*

and $(|\{p\} \cap \bigcup_{k=0}^i \psi_k| - |\{p+1\} \cap \bigcup_{k=0}^{i-1} \psi_k|) \geq 0$. So $\beta - \alpha \geq |\psi_0| - |\psi_i| - 1 \geq 0$.

If ψ_0 contain neither n_p nor n_{p+1} , then $(|\psi_0 \cap \{0, 1, 2, \dots, p\}| - |\psi_0 \cap \{0, 1, 2, \dots, p-1\}|) = 0$ and $(|\{p\} \cap \bigcup_{k=0}^i \psi_k| - |\{p+1\} \cap \bigcup_{k=0}^{i-1} \psi_k|) \geq -1$. Therefore, $\beta - \alpha \geq |\psi_0| - |\psi_i| - 1 \geq 0$.

From the above four discussions, $\beta \geq \alpha$ always holds for any value p . $\square$

Lemma 4.5. *In the scheduling constructed by Algorithm3, the time L_p for each node n_p to transmit all packets for the all-to-all connection is equal to $(w-1)\delta + N - B - 1$ for any p in $0 \leq p \leq N - B - 1$.*

Proof. The proof is similar to Lemma 4.2. $\square$

The scheduling generated by *Algorithm3* have the characteristic that all L_p are identical for $0 \leq p \leq N - B - 1$, so we denote L_P simply as L . Now, we repeat the transmission scheduling constructed by *Algorithm3* until we obtain that any time span of L_{opt} starting from any position of the repeated scheduling is an optimal scheduling. *Algorithm4* will find an optimal scheduling for repeating *Algorithm3*. *Algorithm4* differs from *Algorithm3* in that time t is repeated by $t - L_{opt}$ times if t is greater than L_{opt} .

Algorithm4

- 1: Sort FR-groups in the descending order of group sizes and the sorted order is: $\psi_0, \psi_1, \dots$, and ψ_{w-1} , where $|\psi_0| > |\psi_i|$, and $|\psi_i| \geq |\psi_j|$ if $j \geq i$, $1 \leq i, j \leq w-1$. And Let n_j^i be the j -th node of ψ_i , where $0 \leq j \leq |\psi_i| - 1$.
- 2: For $p = 0$ to $N - B - 1$ do
- 3: $t \leftarrow s_p$
- 4: For $i = 0$ to $w - 1$ do
- 5: For $j = 0$ to $|\psi_i|$ do
- 6: if $(n_p \neq n_j^i)$
- 7: if $(t > L_{opt})$ then $T(n_p, t - L_{opt}) \leftarrow n_j^i$
- 8: else $T(n_p, t) \leftarrow n_j^i$
- 9: $t \leftarrow t + 1$
- 10: $t \leftarrow t + \delta$

Theorem 4.6. *For any values w and δ , the scheduling constructed by Algorithm4 is an optimal scheduling.*

Proof. The proof is similar to Theorem 4.3. $\square$

5 Conclusion

In this paper, we consider B damaged nodes occurring in an N -node WDM star network and the remaining $N - B$ nodes are partitioned into w FR-groups, i.e. $G_0, G_1, \dots$, and G_{w-1} . This is a new direction that consider with damaged or power-off nodes. We present optimal single-hop scheduling algorithms for two different kinds of situations for all-to-all broadcast with arbitrary number of wavelengths w and tuning latency δ . For the case when the transceivers send the packets to all nodes including themselves, the scheduling length found by the scheduling algorithms we proposed meets the lower bound, $\max\{(\max |G_i|)(N - B), w\delta + N_B\}$. For the other case when the transceivers send the packets to all nodes excepting themselves, we present scheduling algorithms to find the optimal scheduling length that comes up with the lower bound $\max\{(\max |G_i|)(N - B - 1), w\delta + N - B - 1\}$.

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